UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

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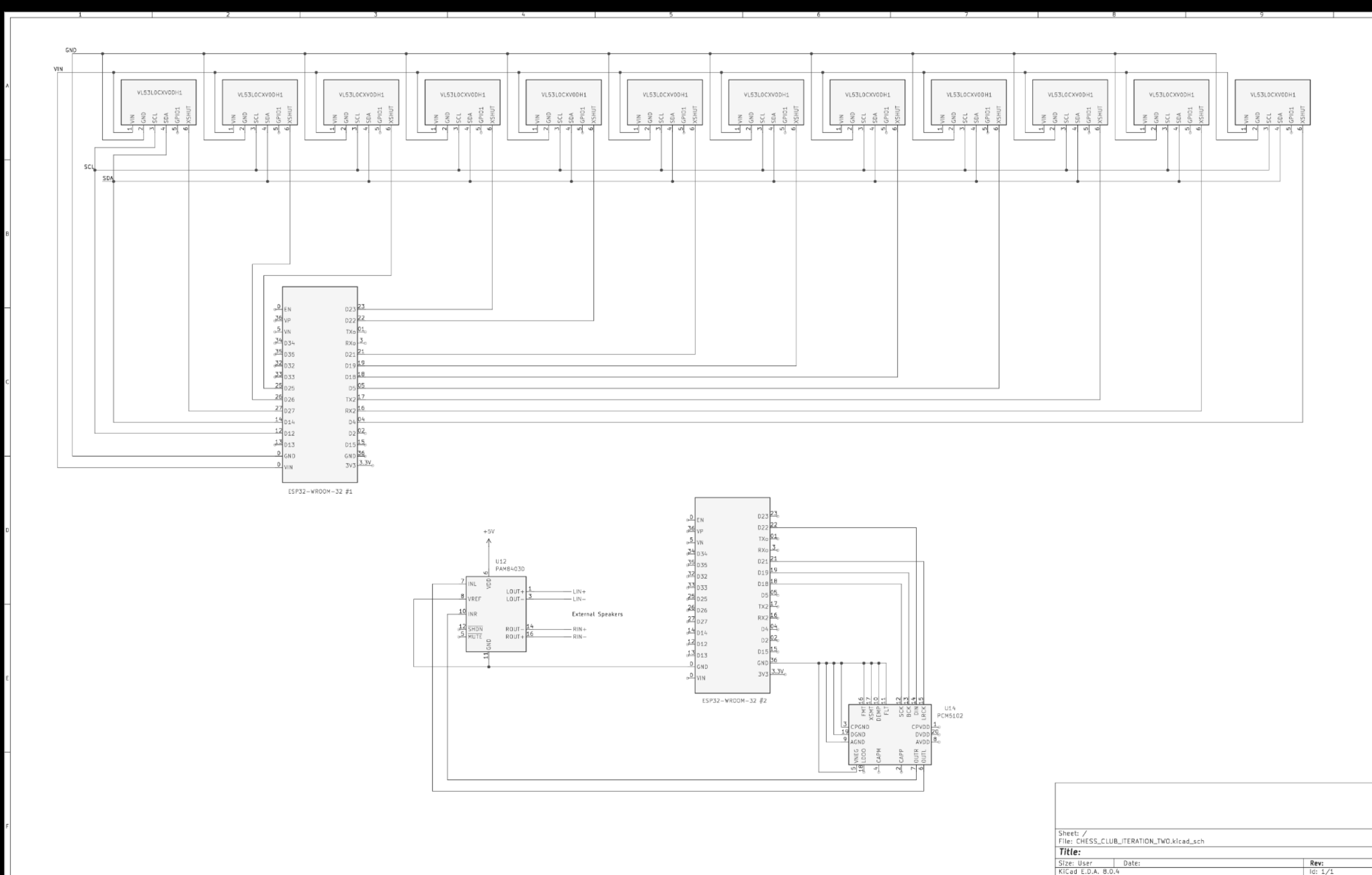
PROJECT OBJECTIVES

- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors.**
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

SCHEMATIC OVERVIEW

** power*
- *3.3V power supply*
    - *VCC + GND*
- 20 power (VDD) terminals to 3.3V
    - 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speaker

** inputs*
- ***sensors***
    - *VIN to 3.3V*
    - *GND to GND*
    - *SCL to ESP32 SCL*
    - *SDA to ESP32 SDA*
    - *GPIO (none)*
    - *XSHUT to unique GPIO on ESP32*
- ***microphone***

** processors*
- *ESP32s*
- *amplifiers*
- *DAC (Digital-to-Analog Converter)*

** outputs*
- *speakers*
- *subwoofer*
- *oled display*

COMPONENT SPECIFICATIONS

(x7) ESP32-ROVER-E
[documentation](https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e_esp32-wrover-ie_datasheet_en.pdf)

- 2 data processors, 3 audio processors, 1 oscilliscope processor, 1 audio output/master coordinator.
- 3.3V
- 8MB PSRAM, 8MB Flash
- Wifi & Bluetooth

(x11) VL53L4CX Time-of-Flight Sensor [listing](https://www.adafruit.com/product/5425)

- 3.3V
- I2C communication
- SCL, SDA, GPIO, XSHUT pins

(x1) MAX9814 microphone

[documentation](https://cdn-shop.adafruit.com/datasheets/MAX9814.pdf)

(x2) MAX98357 Amplifier

[documentation](https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp)

(x1) 128x64 STEMMA QT OLED display

[documentation](https://learn.adafruit.com/monochrome-oled-breakouts)

REFERENCES

A quick development platform for ESP32-A1S digital effect pedal…!

[repository](https://github.com/hamuro80/blackstomp)

- *Taptempo Stereo (Ping-Pong) Delay*
- *Stereo Chorus*
- *BLE Pedal*
- *MIDI Pedal*
- *Mic Mixer*
- *Gain Doubler*
- *Distortion*

An open source, Arduino compatible, ESP32-based Sample Player and Audio Framework.
[repository](https://github.com/marchingband/wvr?tab=readme-ov-file)

microphone and audio processing
[library](https://github.com/sheaivey/ESP32-AudioInI2S)

PCM1794A DAC
[datasheet](https://www.ti.com/lit/ds/symlink/pcm1794a.pdf?ts=1739053305998&ref_url=https%253A%252F%252Fhr.mouser.com%252F)

OPA1612 I/V CONVERTER
[datasheet](https://www.ti.com/lit/ds/symlink/opa1612.pdf?ts=1739154487554&ref_url=https%253A%252F%252Fwww.ti.com%252Fproduct%252FOPA1612)

ALLOCATION
ESP32 Processor #1: Data collection from ToF sensors
Processor #2: Data collection (...)
#3: Waveform synthesizer & basic manipulations
#4: Undecided waveform manipulation
#5: Undecided waveform manipulation
#6: Oscillator generation for OLED display
#7 Audio output processor & mixer for speakers

CURRENT

- Create a schematic for: ESP32 waveform generator > DAC (Digital to Analog Converter) > Subwoofer (TS to RCA) > Monitors (RCA to TS)
  - Create a schematic for: ESP32 waveform generator > DAC (Digital to Analog Converter) > External Master Clock >

- Create a schematic for: Powering systems - Monitors, sub, ESP32s